

Maintenance Tracking & Calibration Management System

Public project brief - sanitized academic capstone summary | Orlando Alvarez

Public summary note

This is a short public version of a team academic capstone. Company-specific records, logos, names, internal contacts, source-code snippets, and original attachments have been removed. Metrics are project evaluation targets or estimates, not production-verified results.

Project Context

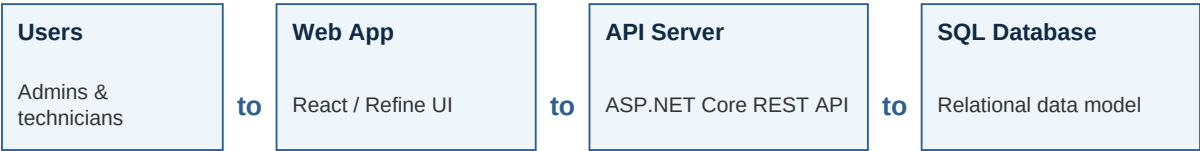
The capstone team designed a web application to move preventive maintenance and calibration tracking away from a manual spreadsheet workflow and into a centralized system.

- Scope included approximately 200 production devices requiring maintenance or calibration tracking.
- Primary users were supervisors/administrators and technicians working with equipment records, task status, and calibration information.
- The project focused on clearer task status, structured forms, maintenance history, dashboards, and KPI-based reporting.

Problem Addressed

- The original workflow used Excel/SharePoint, requiring users to download, edit offline, and re-upload the master file.
- Record updates could take from a few minutes up to three days depending on workload and priority.
- Technicians did not have direct system access and relied on administrative staff for assignments and updates.
- The process created risk around version conflicts, delayed visibility, duplicate/missing equipment IDs, and limited traceability.

System Architecture



Core Features	Technology Stack / Deliverables
Real-time status updates for preventive maintenance and calibration tasks	Frontend: React, Refine, TypeScript
Administrator dashboard for equipment records, task status, and performance indicators	Backend: ASP.NET Core 7.0, Entity Framework, RESTful API
Technician workflow for task assignments, comments, and completion updates	Database: MySQL/PostgreSQL-style relational data model
Structured forms, validation, task history, alerts, and reporting workflows	Docs: C4 diagrams, relational model, quickstart guide, deployment guide

Metrics & Evaluation Plan

The project plan defined measurable targets to check whether the system would improve tracking speed, traceability, and user efficiency.

~200 production devices in scope	90%+ target on-time task completion	< 1 day target registration delay	40-60% target user interaction time reduction	~0% target duplicate/invalid entries
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How the Targets Would Be Measured

Metric	Measurement Approach	Target
Maintenance task punctuality	Compare historical spreadsheet logs with system-based completion logs	At least 90% completed on time
Task registration delay	Compare system timestamps with prior manual entry delays	Reduced to under 1 day
User interaction time	Run time trials with both the spreadsheet workflow and the web workflow	40-60% decrease
Duplicate/invalid entries	Review validation reports and audit logs	Near 0% after validation rules
User satisfaction	Anonymous survey among 5-10 expected users	4.0+ on a 1-5 scale

Engineering Work

• Mapped a manual maintenance/calibration workflow into a full-stack web application concept with frontend, backend, database, and deployment documentation. • Contributed to backend and database design, including relational records for equipment, task status, calibration data, and maintenance history. • Worked with the team to turn stakeholder requirements into screens, API workflows, structured records, and validation rules.	• Applied client-server architecture, REST API design, relational database modeling, and role-based workflows for administrators and technicians. • Documented the architecture, relational model, deployment approach, and evaluation plan so the project could be explained and maintained after delivery. • Kept the public version focused on the engineering problem, implementation approach, and measurable targets without exposing internal project material.
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Public Use Note

This brief uses generalized project context and sanitized details. It does not include company logos, employee names, internal contact information, sensitive operational records, proprietary source code, original attachments, or private file paths.